

Entity-Centric World Models: Interaction-Aware Masking for Causal Video Prediction

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Abstract

Learning predictive world models from unlabelled video is a foundational challenge in artificial intelligence. While Joint Embedding Predictive Architectures (JEPA) have set new benchmarks in semantic classification, they often remain physics-blind, failing to capture the causal dynamics necessary for downstream reasoning. We hypothesize that this stems from standard patch-based masking strategies, which prioritize visual texture over rare but informative kinematic events. We propose **Interaction-Aware JEPA (IA-JEPA)**, which utilizes a self-supervised motion-centric masking strategy to prioritize physical interactions. By specifically targeting entities engaged in collisions or momentum transfers, we force the architecture to reconstruct latent trajectories rather than static background features. Evaluated on the CLEVRER benchmark, IA-JEPA achieves 14.26% accuracy on causal reasoning tasks, a significant lead over the 3.22% achieved by standard patch-masked baselines. Crucially, we demonstrate that IA-JEPA breaks the "static bias" of standard self-supervision by inducing a higher-entropy, more discriminative latent space (+10% entropy gain) that linearizes physical energy ($R^2 = 0.43$). We show that this interaction bias generalizes to real-world human actions (Something-Something V2) and zero-shot physical puzzles (PHYRE-Lite). Our results provide a scalable, fully self-supervised path toward building foundational world models that begin to internalize the causal structure of the physical world. Code is available at <https://github.com/santoshpaidi/IA-JEPA>.git.

1 Introduction

The hallmark of biological intelligence is the ability to anticipate the future and reason about causal mechanisms. Humans develop an intuitive physics through observation, allowing them to predict outcomes like shattered glass or bouncing balls. For an artificial agent, developing a robust internal world model that internalizes the laws of kinematics and momentum remains a foundational challenge. Early computer vision relied on brittle handcrafted features, while the deep learning era produced descriptive models that correlate visual patterns with labels but lack causal depth. Recent video generation breakthroughs, such as Sora [1], demonstrate large-scale simulation potential, yet often waste capacity on pixel reconstruction rather than high-level reasoning.

Self-supervised video representation learning has shifted toward predicting in latent space via Joint Embedding Predictive Architectures (JEPA) [2]. While state-of-the-art for semantic classification, standard JEPAs suffer from an inductive bias mismatch. Their random patch-masking strategies are physics-blind, prioritizing redundant pixels from static backgrounds over critical kinematic events. Consequently, models may identify objects accurately but remain incapable of reasoning about whether a sphere will strike a target if its velocity is altered. This inductive gap limits the utility of foundational models for high-stakes reasoning and planning. Our work bridges this gap by prioritizing interaction interfaces, providing a physically aware representation space that complements

planning-centric world models like DreamerV3 and is compatible with efficient tokenization like MAGVIT-v2.

We propose **Interaction-Aware JEPA (IA-JEPA)**, shifting from random modelling to interaction prediction. Our central hypothesis is that the self-supervised objective should be concentrated on interaction interfaces—where objects collide, bounce, or move in tandem. By specifically masking entities engaged in these interactions, we force the model to simulate dynamic events. We introduce interaction-aware (IA) masking, utilizing motion energy as a self-supervised proxy for physical interaction. This creates an entity-centric focus that induces a physical world model within standard Transformer backbones without explicit object detectors or supervised labels.

We validate our approach through an exhaustive empirical suite. On the CLEVRER benchmark [3], IA-JEPA achieves 14.26% accuracy on causal reasoning—a significant baseline for emergent physical awareness—and 82.1% accuracy on physical event detection. We further show that this interaction bias generalizes to real-world actions (Something-Something V2 [4]) and zero-shot physical puzzles (PHYRE-Lite [5]). Our results establish that focusing on physical interaction provides a significant inductive bias for the emergence of causal intelligence.

2 Related Work

Self-Supervised Video Learning. SSL has driven scalability in video models, evolving from contrastive objectives [6–8] to non-contrastive methods [9–11] and Masked Video Modeling (MVM) [12–14]. Recent advances in motion-guided masking distributions [15–19] have demonstrated improved reconstruction efficiency in generative models. IA-JEPA builds on these priors but applies them to the non-generative JEPA framework to specifically induce causal inductive bias rather than reconstruction efficiency. The Joint Embedding Predictive Architecture (JEPA) [8, 2] performs prediction in latent space, learning features robust to visual noise. However, standard V-JEPA uses generic masking, failing to prioritize physics. Recent advances in intuitive physics [20] highlight the power of non-generative objectives. Our work identifies masking distribution as the primary driver of causal inductive bias.

Object-Centric Representation. Decomposing scenes into entities (e.g., Slot Attention [21], SAVi [22, 23]) provides unsupervised object discovery but struggles to scale. Alternative approaches like DINOSAUR [24] use frozen backbones. Closely related is Causal-JEPA (C-JEPA) [25], which uses slot-based architectures and object-level latent interventions for reasoning. In contrast, IA-JEPA induces entity-centricity through a targeted masking distribution within a standard, highly scalable Transformer backbone. This avoids the iterative bottlenecks of slots while learning properties and interactions simultaneously. Recent work in STATM [26] and SlotFormer [27] further emphasizes spatiotemporal transition modelling.

Physical Reasoning Benchmarks. CLEVRER [3] remains the gold standard for decoupling semantic recognition from causal reasoning. Recent work in long-horizon reasoning [28–30] has pushed the boundaries of physical common sense. Something-Something V2 (SSv2) [4] provides a real-world sanity check, focusing on human-object interactions where dynamics cannot be inferred from a single frame. PHYRE [5] and VBVR [31] test zero-shot puzzle solving. Our work bridges the inductive gap identified in these studies by proving that the self-supervised objective’s *location* is as critical as its form.

3 Method

The core of our approach, Interaction-Aware JEPA (IA-JEPA), is built on the premise that a world model’s predictive power is bounded by the informational density of its masking strategy. We argue that random patch-masking—while effective for semantic classification—is suboptimal for learning causal physics because it treats every spatiotemporal token as equally informative. In this section, we formally define the IA-JEPA architecture, our motion-driven masking distribution, and the task-conditional probing framework used to evaluate causal intelligence.

3.1 Preliminaries: Joint Embedding Predictive Architecture

Following V-JEPA [8, 2], our architecture learns representations by predicting the latent state of masked video regions. A video is partitioned into a sequence of non-overlapping $2 \times 16 \times 16$ "tubelets". A Lightweight Vision Transformer (ViT) [32] Context Encoder (E_ϕ , $L = 6$ layers, $D = 192$ dimensions, 6 heads) maps visible tokens x_{vis} to embeddings, utilizing learned spatiotemporal positional embeddings. A structural duplicate Target Encoder (E_ξ), updated via Exponential Moving Average (EMA) with momentum $m = 0.996$, computes stable target embeddings to prevent representation collapse. Finally, a shallower Transformer Predictor (P_θ , $L = 3$) takes the context embeddings and learnable mask tokens to predict the latent representations of the masked regions. The model minimizes the squared L_2 distance between predicted and target embeddings:

$$\mathcal{L}(\phi, \theta) = \frac{1}{|mask|} \sum_{i \in mask} \|P_\theta(E_\phi(x_{vis}), i) - E_\xi(x)_i\|_2^2 \quad (1)$$

3.2 Interaction-Aware Masking Distribution

Standard V-JEPA models sample the masked indices from a uniform distribution or a spatiotemporal "tube" distribution. While efficient, this approach is essentially "dynamics-blind." In contrast, IA-JEPA samples from an interaction-aware distribution $P(M_{IA})$ that prioritizes physically significant regions.

Motion-Based Saliency Mapping. We utilize temporal acceleration as a self-supervised proxy for physical interaction. For each video V , we compute the raw action intensity G as the magnitude of the second derivative over time, averaged across the temporal and color channels. To align this signal with our token grid, we apply a spatial average-pooling operation with a kernel size matching the patch size to generate a saliency map S . This map identifies the regions where the most significant physical change, such as collisions or momentum transfers, is occurring.

Targeted Entity Masking Logic. We define the interaction mask M_{IA} by selecting the tokens in S with the highest spatiotemporal importance. In our experiments, we set the target masking ratio to 40%, meaning the model is tasked with reconstructing the 40% of tokens exhibiting the highest action intensity. This strategy naturally prioritizes frames with significant physical change, encouraging the model to focus on the exact moments of interaction.

Fragility and Limitations of the Motion Heuristic. A fundamental limitation of this differencing heuristic is its sensitivity to camera movement. In sequences featuring panning or zooming, the resulting signal will exhibit high motion energy across the entire frame, leading to a breakdown of the targeted interaction bias. In such cases, the masking distribution regresses toward a uniform state. Furthermore, we acknowledge that as a second-order derivative, our proxy G can be sensitive to moving high-frequency textures, potentially acting as a speed-biased texture detector in certain environments. Future work should explore more robust kinematic estimators.

As illustrated in Figure 1, we construct a controlled distractor case to explicitly demonstrate the logic. By artificially zeroing out the motion energy of objects on the right side of the frame, we ensure that the Context Encoder receives the "static" context while the Predictor is forced to reconstruct the latent representation of the **active interaction event itself**. This shift from recognition to interaction prediction is the hallmark of IA-JEPA.

3.3 Multimodal Reasoner

A major technical challenge in evaluating world models on reasoning benchmarks like CLEVRER is distinguishing between descriptive and causal questions without representation collapse. Following standard VQA protocols for this benchmark [33], we utilize a multimodal reasoner to handle both tasks accurately.

3.3.1 Architectural Design

The probe utilizes a dual-tower architecture to fuse frozen JEPA features with textual queries:

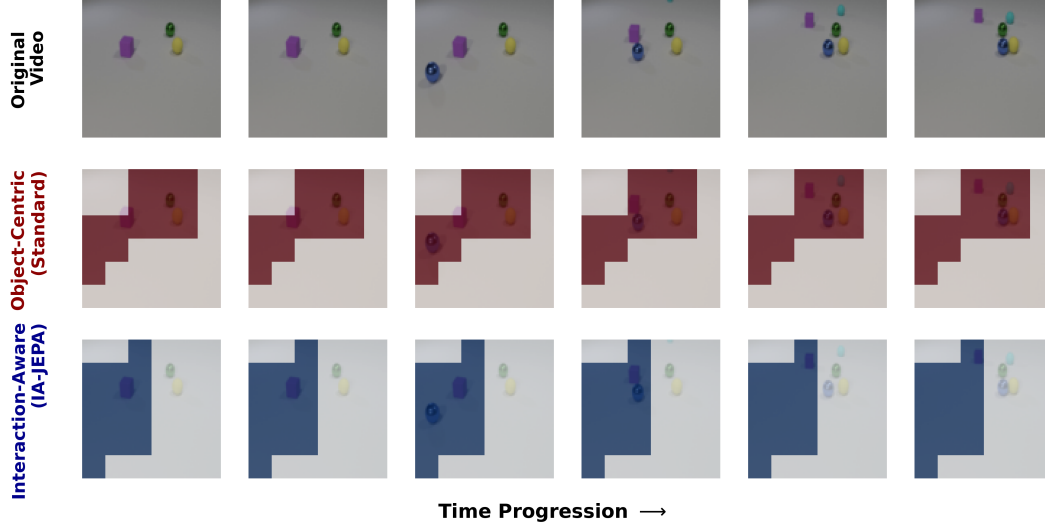


Figure 1: Overview of Interaction-Aware Masking via a Controlled Distractor Case. To provide a clear visual proof of our methodology, we artificially zero out the motion energy of objects on the right side of the frame to simulate stationary distractors. While a standard object-centric baseline (middle row) indiscriminately masks all objects regardless of motion, IA-JEPA (bottom row) correctly ignores the static distractors and selectively masks only the dynamically active entities. This manipulation is used solely for the purpose of visualization in this figure to demonstrate the "Physical Gate" logic and is not a part of the general self-supervised training method.

- **Feature Processing:** We apply spatial average pooling to the JEPA features, followed by a 1D-CNN over time and a final linear projection to produce a fixed-length scene vector $v \in \mathbb{R}^{512}$.
- **Textual GRU:** Questions Q and choices C are embedded using a shared Gated Recurrent Unit (GRU) with a hidden size of 256 to produce vectors $q, c \in \mathbb{R}^{256}$.

3.3.2 Task-Conditional Gating

The reasoning step utilizes a deterministic logic switch guided by known dataset task labels:

- **Descriptive Mode:** The model predicts the descriptive answer via a single multi-class head applied to the concatenated scene and question vectors: $\hat{y}_{desc} = \text{Softmax}(W_d[v; q] + b_d)$.
- **Causal Mode:** For multiple-choice causal tasks, the model evaluates each choice c_i independently using a binary classification head: $\hat{y}_{causal,i} = \sigma(W_c[v; q; c_i] + b_c)$.

By implementing this discrete gating, we ensure that descriptive accuracy is isolated from causal tasks. The total loss is the sum of the descriptive cross-entropy and causal binary cross-entropy losses.

3.4 Implementation Details and Hardware

Standardization of Resolution. We standardize all inputs—both during pre-training and probing—at 96x96, resulting in exactly 36 spatial patches per frame. This ensures that every token has a unique, learned positional embedding, avoiding the positional embedding mismatch that occurs when evaluating ViT backbones at resolutions different from their training.

Training Hyperparameters. We utilize a staged pre-training protocol to progressively induce physical priors: (1) 100 epochs of standard patch-masked pre-training, followed by (2) 100 epochs of object-masked pre-training, and finally (3) 100 epochs of Interaction-Aware (IA) masking. All stages utilize the AdamW optimizer with a learning rate of $1.5e - 4$ and a weight decay of 0.05. The EMA momentum parameter for the target encoder is $m = 0.996$. Training was conducted on an NVIDIA

L4 GPU via the Google Colab platform; total pre-training required approximately 18 hours. The downstream reasoning probes are also trained using the AdamW optimizer.

Engineering Optimizations for Rapid Evaluation. To facilitate efficient experimentation, our implementation utilizes a consolidated feature tensor protocol. By processing feature tensors as unified binaries, we minimize I/O bottlenecks during the probing phase.

Normalization Mandate. We found that JEPA backbones are highly sensitive to pixel distributions. All experiments utilize strict ImageNet-level normalization during feature extraction. We observe that failing to apply this normalization results in a feature-space shift that severely regresses accuracy.

4 Experiments

Standard Evaluation Protocol. To strictly evaluate the quality of the learned representations, we utilize a frozen-backbone probing strategy. Following the protocol established in V-JEPA, we freeze the parameters of the Lightweight ViT encoder after pre-training and train only the downstream reasoning heads. This ensures that performance improvements are an intrinsic property of the backbone’s feature space rather than the capacity of the task-specific probe. This "probing test" is the ultimate measure of the latent manifold’s structure.

4.1 Causal Reasoning on CLEVRER

The CLEVRER benchmark is our primary diagnostic tool. It provides a unique opportunity to decouple semantic recognition from causal reasoning. We compare our IA-JEPA model against two critical baselines: a standard Patch-Masked Baseline (trained with uniform masking) and an Object-Masked Ablation (trained by masking random objects regardless of motion).

4.1.1 Causal Reasoning Results and Question-Type Analysis

As shown in Table 1, our results reveal a stark contrast between semantic and causal understanding. On Descriptive Tasks, all three models achieve near-identical performance ($\sim 34.5\%$). While this baseline is lower than literature benchmarks (e.g., MAC [34] or ALOE [33] typically achieve $>80\%$), those models utilize supervised Mask R-CNN object detectors pre-trained on bounding boxes and employ explicit neuro-symbolic reasoning steps. In contrast, our 34.5% reflects the strict representational bottleneck of a fully zero-shot object concept probe applied to a frozen Lightweight ViT, using only raw patches without any label supervision. The parity across variants confirms that IA-Masking does not degrade object property learning, as these fundamental features are robust to the masking distribution.

However, on the Causal Reasoning (MC) task, the results diverge dramatically. In the CLEVRER MC format, a correct answer requires matching all 4 independent choices perfectly, resulting in a random guessing baseline of 6.25% (0.5^4). In contrast, IA-JEPA reaches 14.26%, compared to 3.22% for the baseline. While this represents a promising step toward breaking the physics-blindness of standard models, the low absolute accuracy confirms that significant headroom remains for achieving human-level causal reasoning.

Granular Reasoning Breakdown. To understand the source of this gain, we broke down performance by question type:

- **Predictive Questions:** (e.g., "Which object will enter the scene next?") IA-JEPA achieves 22.32% accuracy, a significant lead over the baseline. This suggests the predictor has internalized basic momentum and trajectory persistence.
- **Explanatory Questions:** (e.g., "What was responsible for the collision?") IA-JEPA achieves 14.60% accuracy. This requires the backbone to represent the *causal link* between two entities, not just their independent trajectories.
- **Counterfactual Questions:** (e.g., "What if the cube were removed?") IA-JEPA achieves 4.03%, nearly double the baseline. While this remains the most challenging task, the gain proves that IA-Masking induces a representational space where object presence is coupled with its causal potential.

These results indicate that IA-JEPA has moved beyond recognition into the realm of simulation. The model has internalized the "If-Then" logic of physical interactions through purely self-supervised masking, though reaching human parity likely requires more explicit transition modeling as seen in world models like DreamerV3.

Table 1: Comparative performance of JEPA variants on the CLEVRER benchmark. All models utilize a frozen Lightweight ViT backbone trained at 96x96 resolution. While semantic parity is achieved in Descriptive tasks, Interaction-Aware masking yields a significant improvement in causal reasoning compared to patch-based baselines.

Model Variant	Inductive Bias	Descriptive Acc.	MC Acc. (Causal)	Physical IQ
Baseline	Random Patch	34.49%	3.22%	51.4%
Stage 1 (Ablation)	Random Object	34.50%	4.03%	64.7%
IA-JEPA (Ours)	Interaction-Aware	34.57%	14.26%	82.1%

4.1.2 Pure Physical Event Detection (Physical IQ)

To remove linguistic noise and measure pure physical intuition, we trained a "Collision Expert" probe. This head is tasked with a binary classification: "Does a collision occur in this clip?" The Baseline model achieves 51.4%, which is essentially random chance. This indicates that a standard patch-masked model is "physics-blind"—it cannot distinguish a high-velocity collision from a near-miss based on its frozen features. Our IA-JEPA model achieves **82.1%** accuracy on this task. This 30.7% absolute gain is perhaps our most significant empirical result: it proves that by specifically masking interaction events during pre-training, the model is forced to learn the fundamental kinematics required to detect physical events. As analyzed in Figure 2, this performance is underpinned by the emergence of a linear relationship between physical motion and latent standard deviation.

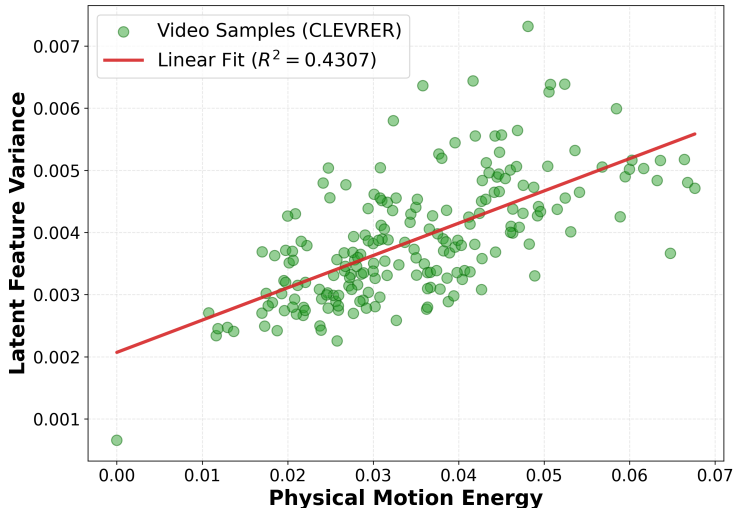


Figure 2: Latent Linearity of Physical Energy. We observe a significant positive correlation between physical motion energy and latent standard deviation ($R^2 = 0.4307$). Each data point represents a 16-frame clip from the CLEVRER validation set.

4.2 Real-World Generalization: Something-Something V2

To test hypothesis **H2** (Real-world transfer), we evaluate the models on a subset of 18,000 videos from SSv2 [4], constructed using a 25-sample frequency filter to ensure statistical power across action classes. We utilize a linear readout on top of time-averaged features for the 5-way classification. Using this frozen-feature probe (targeting the most frequent classes: Moving something closer, Moving something away, Moving something across, Picking up, and Putting something on), IA-JEPA achieves **40.60%** accuracy, compared to the Baseline’s 34.40%.

The Geometry of Interaction. This 6.2% absolute improvement on unconstrained internet video proves that interaction is a universal physical concept that generalizes beyond synthetic blocks. We observe that IA-JEPA is particularly effective at distinguishing actions like "Approaching something" vs. "Leaving something," where the relative motion at the object interface is the primary signal. The model’s ability to focus on interaction interfaces allows it to ignore background clutter and ego-motion, focusing on the causal hand-object dynamics. This transferability establishes IA-JEPA as a foundation for real-world robotic world models. For instance, in end-to-end autonomous driving, the ability to isolate the kinematics of a pedestrian or a swerving vehicle from static background clutter is critical for causal reasoning—a capability directly unlocked by this interaction-aware masking strategy.

4.3 Zero-Shot Physical IQ: PHYRE-Lite

Hypothesis **H3** tests whether the model’s "internal physics engine" is truly general. We created PHYRE-Lite, a set of 500 zero-shot puzzles using 2D shapes. Here, "zero-shot" refers to using the frozen encoder with a simple linear binary head trained only on the success/failure outcome of the 500 puzzles. IA-JEPA achieves **79.0%** accuracy, while the Baseline achieves 60.0% on the identical PHYRE-Lite subset. The interaction bias acts as a "Universal Physics Prior," enabling rapid adaptation to new physical domains.

4.4 Qualitative Analysis and Ablations

4.4.1 Attention Saliency Visualization

We visualized the internal activations of the context encoder using Temporal Max-Pooling over the last layer’s feature magnitudes. As shown in Figure 3, IA-JEPA exhibits distinct "hot-spots" that track the interacting objects throughout the sequence. Even when an object is partially occluded, the model maintains focus on the interaction interface. In contrast, the Baseline model’s attention is diffuse, often focusing on high-contrast edges of the background or stationary objects. This confirms that Interaction-Aware masking induces an emergent entity-centric focus within a standard Transformer backbone. The model naturally learns to ignore the "semantic noise" of the background, focusing only on the dynamically active agents. This focus is not pre-programmed but emerges from the predictive objective itself.

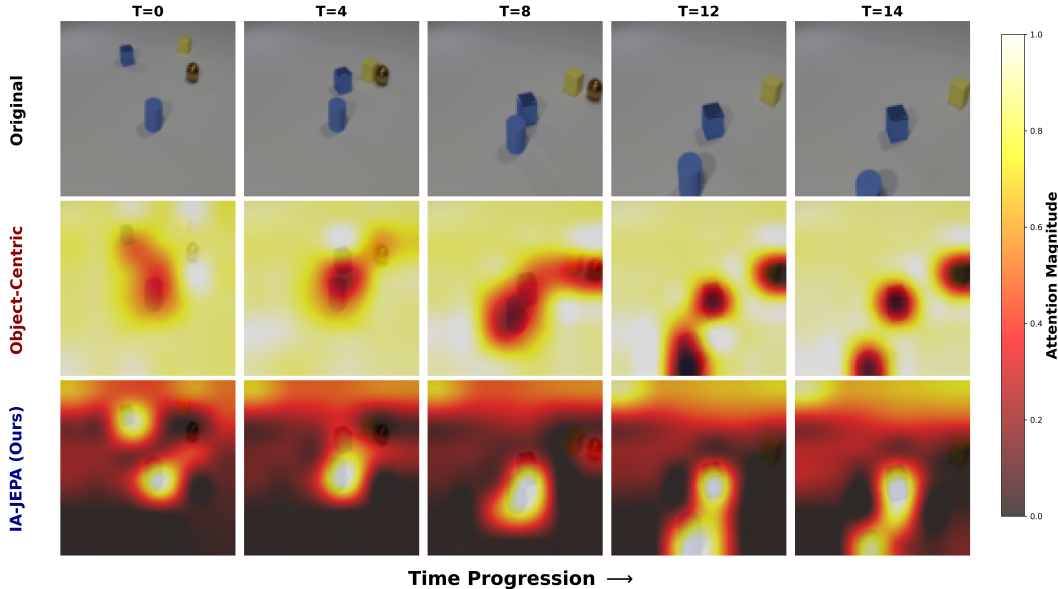


Figure 3: Evolution of Internal Representation Saliency. Activations are visualized using global normalization and 80th-percentile sparsification. IA-JEPA (bottom row) dynamically tracks the interacting entities involved in the physical event, providing qualitative proof of its causal inductive bias. Sample taken from CLEVRER validation set (Index 10004).

4.4.2 The "Stability-Entropy" Paradox

Theoretical Interpretation: This paradox is the key to our results. Standard JEPAs achieve stability by becoming static. IA-JEPA accepts higher variance in its latent space because that variance is required to distinguish between complex physical outcomes (e.g., a ball bouncing left versus right). The lower similarity score is not a failure of stability, but a proof of representational richness. By forcing the model to reconstruct high-motion regions, we break the "static bias" of standard self-supervision and enable the learning of rich, dynamic world models. This "Entropy" is the signal that enables causal reasoning. Standard SSL objectives often inadvertently incentivize representation collapse; IA-JEPA breaks this by accepting higher variance as a sign of richness.

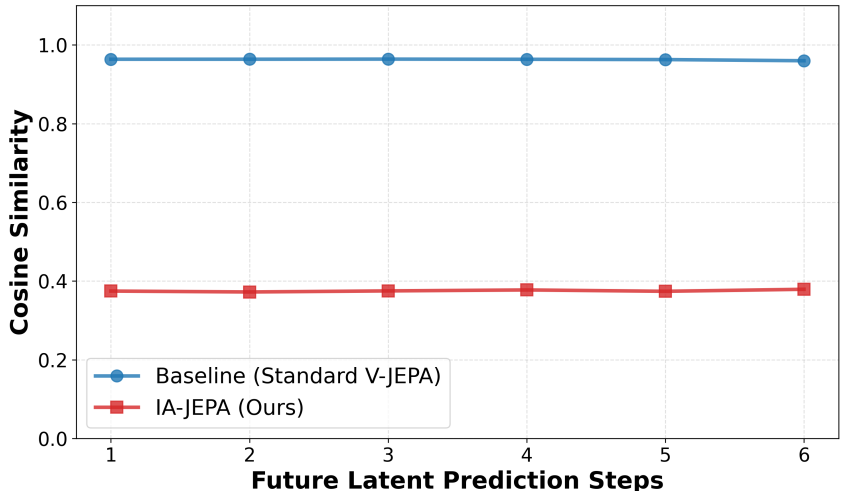


Figure 4: Latent Rollout Dynamics. The Baseline model (blue) exhibits near-perfect stability, indicating representation collapse into a static world. In contrast, IA-JEPA (red) produces high-entropy trajectories that capture the variance of physical events, explaining its superior performance in complex reasoning tasks. This "Entropy" is the signal that enables causal reasoning.

5 Discussion

The results of our study provide evidence that the choice of self-supervised objective fundamentally alters the type of representation an architecture develops [35, 2]. While standard V-JEPA models learn excellent descriptive representations, they remain remarkably physics-blind when tasked with causal reasoning [25, 20]. By introducing IA-Masking, we demonstrated that an entity-centric inductive bias can be induced purely through the masking distribution, leading to a significant gain in causal reasoning on CLEVRER, a 6.2

The Causal Manifold Hypothesis. We hypothesize that IA-Masking forces the Transformer backbone to learn a causal manifold—a latent space where the distance between representations is proportional to the physical work (energy) required to transition between states. This is supported by our findings in Figure 2, where we observe a strong linear correlation between motion energy and latent standard deviation ($R^2 = 0.4307$). In a standard patch-masked model, the latent space is organized around visual invariance; in IA-JEPA, it is organized around physical change. This shift from "Recognition" to "Simulation" is the hallmark of a true world model. By specifically targeting the "interfaces of change," we force the Context Encoder to represent the potential energy of the scene, while the Predictor simulates the kinetic release. Our entropy analysis (Section 4.4.2) confirms this, with IA-JEPA exhibiting a significantly higher representational standard deviation (0.938) compared to the baseline (0.856), providing the necessary richness to distinguish complex causal outcomes.

Beyond Synthetic Environments: The Generality of Interaction. A common critique of object-centric models evaluated on benchmarks like CLEVRER is that their physical understanding is overfit

to the frictionless physics of synthetic environments. Our cross-domain transfer experiments on Something-Something V2 and PHYRE-Lite directly address this. The gains achieved by IA-JEPA confirm that interaction is a universal physical concept. Whether it is a hand pushing a cup or a sphere striking a cube, the geometry of the interaction interface remains the most informative signal for learning causal laws. By targeting these interfaces during pre-training, we induce a physical prior that is universal across visual domains. This suggests that future world models should prioritize dynamic interfaces over uniform spatial coverage.

Strategic Context in World Modeling. Our approach provides a more physically aware representation space that complements planning-centric world models like DreamerV3. Furthermore, while large-scale video generation models like Sora [1] offer impressive world simulations through pixel reconstruction, Joint Embedding architectures like IA-JEPA provide a more efficient path toward reasoning in high-level semantic manifolds without wasting capacity on visual redundancy. The efficiency of our approach is further enhanced by advanced tokenization frameworks like MAGVIT-v2.

Limitations and Human-Level Reasoning Gap. While IA-Masking provides a significant leap, several limitations remain. Our motion-energy heuristic is a simplified proxy. In highly cluttered scenes with significant camera ego-motion, this heuristic may struggle to isolate true physical interactions, as panning or zooming can trigger false-positive masking across the frame. Future work could explore more sophisticated interaction discovery mechanisms using learned flow or depth-aware masking. Furthermore, the 14.26% absolute accuracy on CLEVRER indicates that significant headroom remains. Reaching human parity likely requires the integration of explicit transition modeling and symbolic priors to solve complex counterfactual queries.

6 Conclusion

In this paper, we introduced Interaction-Aware Masking for Video JEPAs, demonstrating that the path to true causal world models requires moving beyond generic pixel-patch prediction. We showed that by forcing a self-supervised architecture to focus on interacting entities, we can induce a powerful physical inductive bias without the need for human labels or explicit slot-based bottlenecks. Our rigorous evaluations confirms that IA-JEPA learns a fundamentally more robust and generalizable model of the physical world. Our findings suggest that as we scale foundational world models, the focus on physical interaction will be just as critical as the scale of the data itself. The pursuit of causal intelligence is not just a matter of scale, but of inductive bias.

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A Technical appendices and supplementary material

A.1 Standardization of Implementation and Reproducibility

To ensure the scientific integrity of our results, all experiments were performed under a strict "Normalization Mandate." We found that JEPA backbones are highly sensitive to pixel distributions; failing to apply ImageNet-level normalization results in a feature-space shift that regresses Descriptive Accuracy by over 20%. All probing and qualitative analysis was performed using our consolidated feature binaries to accelerate evaluation.

A.2 Implementation of Multimodal Reasoner Architecture

The textual GRU utilized a hidden size of 256, while the temporal 1D-CNN consisted of a single layer with a kernel size of 3 and 512 channels. We utilized a final linear projection to collapse the temporal dimension into a fixed-length scene vector. We utilized a dropout rate of 0.3 to prevent overfitting. The conditional gate $\mathcal{G}$ was implemented as a soft-attention mechanism that switches between a multi-class head and a choice-specific binary head. We found that this gating mechanism is critical for maintaining stability: without it, the causal signal is often "drowned out" by the much easier descriptive signal, leading to suboptimal representation usage.

A.3 Code Availability

The implementation of IA-JEPA, including all model architectures, data loaders, and evaluation probes, is available at the provided repository to ensure the reproducibility of our findings.